

Coding & Solution

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This document summarizes the implemented EduGenie solution: where the code is, what it is built with, which features work, what is still pending, and how to run it.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/bhavana-cse/EduGenie Live app: https://edu-genie-mauve.vercel.app
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	FastAPI, Uvicorn, Jinja2; Google GenAI SDK, Hugging Face Transformers, PyTorch
Key Features Implemented	• Question Answering: short, accurate answers from the Google Gemini API. • Concept Explanation: simple explanations from the local LaMini-Flan-T5 model, with automatic fallback to Gemini. • Quiz Generator: 3 multiple-choice questions with 4 options, checked for valid JSON, with instant Correct / Incorrect feedback. • Text Summarization: long passages condensed into short, simple notes. • Learning Path: beginner, intermediate and advanced plan with timelines and resources. • Responsive single-page interface with a task dropdown and readable formatted results. • Clear error messages, and the API key kept in environment variables.
Pending / Incomplete Features	• User login and per-user history (the first release is open access). • Progress dashboard with past quiz scores. • Upload of PDF or image files instead of typed text. • Voice input and multilingual support. • Automated unit tests.

Setup / Run Instructions

1. Clone the repository from GitHub and open the folder in Visual Studio Code.
2. Create and activate a virtual environment (python -m venv .venv, then .venv\Scripts\activate on Windows).
3. Install the dependencies: pip install -r requirements.txt
4. Copy .env.example to .env and set GEMINI_API_KEY (from Google AI Studio) and GEMINI_MODEL.
5. Start the server: uvicorn main:app --reload
6. Open <http://127.0.0.1:8000> in a browser and choose a task from the dropdown.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions and modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments and documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository (GitHub)	Yes

Additional Notes / Comments

The originally planned Gemini 1.5 Pro model has been retired by Google, so the model name is read from the GEMINI_MODEL setting and can be changed without editing any code. The first concept explanation downloads the local LaMini model (about 3 GB) once; if the model cannot load, the app uses Gemini so the feature still works.